

REVOFIN

Fintech Lending Platform

Loan Portfolio Analysis

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Data Cleaning & Standardization

Preparing a clean, analysis-ready dataset from raw BigQuery tables

```
-- Customer table: strip non-alphanumeric chars from customer_id
SELECT
```

```
  array_to_string(
    regexp_extract_all(customer_id, '[a-zA-Z0-9]'), ''
  ) AS customer_id_clean,
  annual_inc, home_ownership,
  emp_length, addr_state
FROM `nextproject-001.FIP.customer`
```

```
-- Loan table: same cleaning + status reclassification
SELECT
```

```
  array_to_string(
    regexp_extract_all(loan_id, '[a-zA-Z0-9]'), ''
  ) AS loan_id_clean,
  array_to_string(
    regexp_extract_all(customer_id, '[a-zA-Z0-9]'), ''
  ) AS customer_id_clean,
  funded_amount, int_rate, loan_status,
  issue_year, purpose, addr_state,
  -- Reclassify loan_status
CASE
  WHEN loan_status = 'Does not meet the credit policy.'
  THEN 'Current'
  WHEN loan_status = 'In Grace Period'
  THEN 'In Grace Period'
  ELSE loan_status
END AS loan_status_clean
FROM `nextproject-001.FIP.loan`
```

CLEANING RESULTS

✓ Customer table: customer_id normalized via regexp_extract_all — alphanumeric characters only retained

✓ Loan table: loan_id & customer_id cleaned using identical regex pattern for consistent join keys

✓ Loan status reclassified: 'Does not meet credit policy' → Current; duplicate categories merged

✓ int_rate stored as decimal (0–1); multiplied ×100 for all percentage displays in this report

✓ funded_amount used as Outstanding Balance (OS) proxy — no out_prncp column available in dataset

⚠ issue_year stored as FLOAT64; cast to INT64 for cohort grouping and year-level aggregation

Portfolio Filtering & Scope Definition

Isolating the active loan population for analysis

```
SELECT *  
FROM `nextproject-001.FIP.loan`  
WHERE loan_status_clean IN (  
  'Fully Paid',  
  'Current',  
  'In Grace Period',  
  'Late (31-120 days)',  
  'Default'  
)
```

LOAN STATUS CLASSIFICATION

Fully Paid	58.3%	Completed loans — no active risk
Current	37.4%	Performing within scheduled terms
In Grace Period	1.2%	≤15 days past due — elevated attention
Late (31-120 days)	2.1%	DPD > 30 — contributes to TKB30 calc
Default	1.0%	Written-off — excluded from ENR

176,087

Total Loans
in active portfolio

\$2.80B

Portfolio OS
funded_amount basis

\$2.80B

ENR Balance
excl. Default loans

98.11%

TKB30
portfolio quality

Scope Definition

- All five loan status categories are retained to enable complete delinquency analysis.
- Late (31-120 days) and Default loans are included in TKB30 calculation as DPD>30 population.
- ENR excludes Default balances per standard fintech risk accounting practice.

Executive Summary

Key metrics snapshot as of portfolio cut-off date

176,087

Total Active Loans

Fully Paid + Current + IGP + Late +
Default

\$2.80B

Outstanding Balance

Sum of funded_amount across
portfolio

\$2.80B

Eligible Net Revenue

ENR: portfolio excl. Default loan
balances

98.11%

TKB30 Quality Score

Portfolio not DPD>30 / total portfolio
OS

METRIC DEFINITIONS & FORMULAE

Outstanding Balance (OS)

$\Sigma \text{ funded_amount}$ for all active loans

funded_amount used as OS proxy; no out_prncp column present in dataset

Eligible Net Revenue (ENR)

$\Sigma \text{ funded_amount}$ WHERE loan_status $\neq$ 'Default'

Represents collectible balance — excludes written-off exposure from Default category

TKB30 (Portfolio Quality Score)

$1 - (\text{OS DPD}>30 / \text{Total OS}) \times 100\%$

DPD>30 = Late (31–120 days) + Default; TKB30 result: 98.11% → healthy portfolio

TKB30 Portfolio Quality Score

Delinquency-based portfolio health metric — overall portfolio assessment

TKB30

Portfolio Quality Score

98.11%

EXCELLENT

Total OS \$2,800M

DPD>30 OS \$53M

% at Risk 1.89%

TKB30 Formula

$$\text{TKB30} = 1 - (\text{OS_DPD30} / \text{OS_Total}) \times 100\%$$

OS_DPD30 = Σ funded_amount WHERE status IN ('Late (31-120 days)', 'Default')

OS_Total = Σ funded_amount (all statuses)

Result → TKB30 = 98.11%

DPD>30 POPULATION BREAKDOWN

Loan Status	OS	OS %	DPD>30
Fully Paid	\$1,631M	58.25%	No
Current	\$1,047M	37.39%	No
In Grace Period	\$33M	1.18%	No
Late (31-120 days)	\$31M	1.11%	YES
Default	\$22M	0.79%	YES

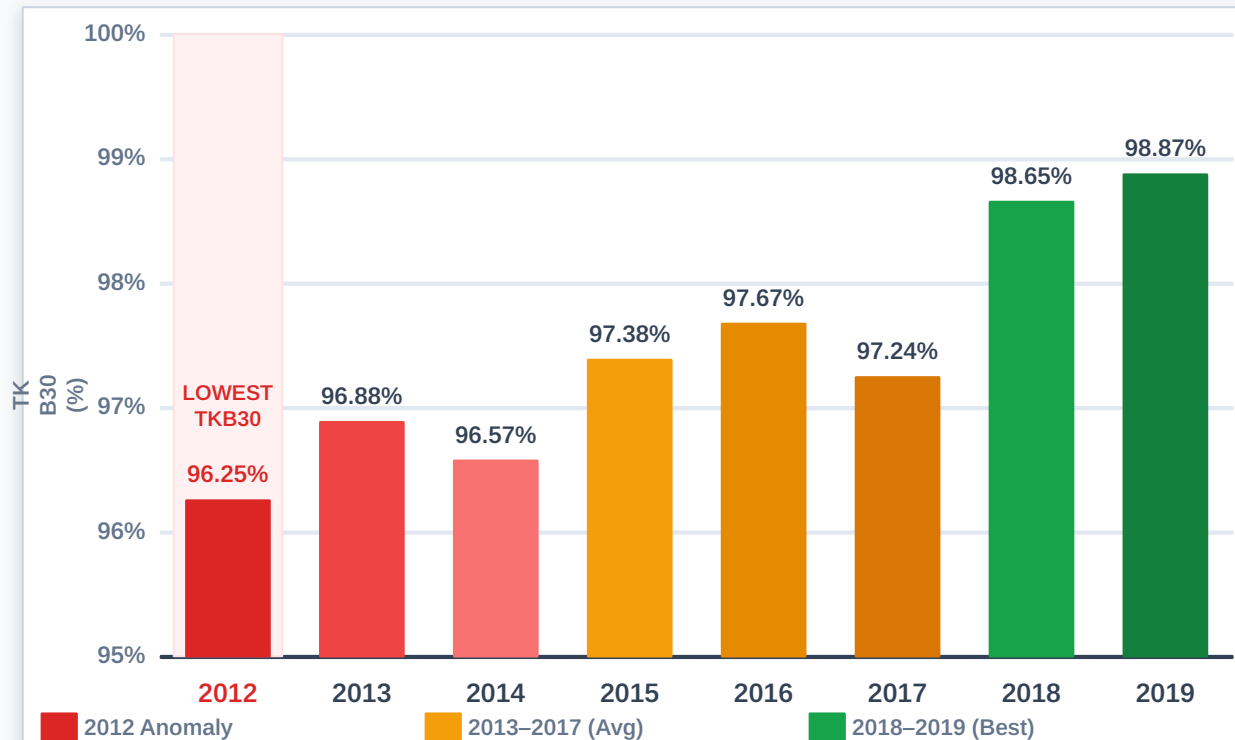
Cohort Analysis by Issuance Year

Performance across 2012–2019 loan cohorts

Cohort (Yr)	Loans (#)	OS (\$M)	DPD>30 (\$M)	TKB30 (%)	Avg Rate (%)	Assessment	
2012	1,218	\$21.0	\$0.79	96.25%	18.36%	Anomaly	
2013	4,183	\$73.7	\$2.30	96.88%	15.20%	Below Avg	
2014	10,249	\$184.3	\$6.32	96.57%	14.90%	Below Avg	
2015	22,638	\$416.8	\$10.91	97.38%	13.40%	Average	
2016	31,574	\$580.4	\$13.51	97.67%	13.10%	Average	
2017	43,018	\$770.2	\$21.25	97.24%	12.70%	Average	
2018	37,605	\$544.3	\$7.33	98.65%	12.20%	Good	
2019	25,602	\$209.3	\$2.36	98.87%	12.00%	Best	

TKB30 Trend & Anomaly Identification

Cohort TKB30 performance across 2012–2019 — detecting outlier vintages



KEY FINDINGS

⚠️ 2012 — Worst Cohort

TKB30: 96.25% — lowest across all vintages. Flagged as anomaly requiring root cause investigation.

↗️ 2019 — Best Cohort

TKB30: 98.87% — highest quality vintage. Strong loan screening and favorable market conditions.

→ Clear Upward Trend

Steady improvement from 96.25% (2012) to 98.87% (2019), reflecting tightening underwriting standards.

~ 2017 Minor Dip

TKB30: 97.24% — slight dip within 2015–2017 band; warrants monitoring but not anomalous.

Anomaly Deep-Dive: Demographic Factors

Do customer demographics explain the 2012 cohort's poor performance?

HYPOTHESIS: Lower-quality borrowers (lower income, weaker employment, higher-risk home ownership) drove the 2012 anomaly.

△ Home Ownership

MORTGAGE HOLDERS

2012 Cohort: 52.68%

Others Avg: 49.04%

+3.64 pp

▲ BETTER

2012 borrowers had MORE mortgage holders — a lower-risk indicator. Demographics do NOT explain anomaly.

◎ Employment Length

AVG. EMPLOYMENT (YEARS)

2012 Cohort: 6.59 yrs

Others Avg: 5.85 yrs

+0.74 yrs

▲ BETTER

2012 borrowers had LONGER employment tenure — indicating more stable income. No negative demographic signal.

\$ Annual Income

MEDIAN ANNUAL INCOME

2012 Cohort: \$69,000

Others Avg: \$67,000

+\$2,000

▲ BETTER

2012 borrowers earned MORE income on average. Income does not explain higher delinquency in this cohort.

✓ **VERDICT:** Demographic factors are NOT the cause of the 2012 anomaly. 2012 borrowers had better income, employment, and home ownership than other cohorts.

Financial Driver Analysis: Root Cause Identified

Interest rate and loan purpose explain the 2012 cohort anomaly

ROOT CAUSE: 2012 cohort carried significantly higher interest rates (+5.43 pp) and a greater share of debt consolidation loans (+9.14 pp) — two high-risk signals.

TASK 7d — TOP STATE: CALIFORNIA (CA)

CA share in 2012 cohort	15.18%	Highest single-state concentration
CA loans in 2012 cohort	185	out of 1,218 total 2012 loans
CA cohort TKB30 (2012)	96.10%	Slightly below cohort average of 96.25%
Implication	—	State mix is NOT primary driver; CA pattern mirrors the broader cohort

TASK 7e — DOMINANT PURPOSE: DEBT CONSOLIDATION



TASK 7f — PRIMARY DRIVER: INTEREST RATE (+5.43 pp GAP)

2012 Cohort (Anomaly)

18.36%

Average interest rate

HIGH RISK ►

Other Cohorts (Avg)

12.93%

Average interest rate

BASELINE

Conclusions & Strategic Recommendations

Key findings and actionable risk management framework

KEY CONCLUSIONS

C1

Portfolio Quality is Strong

Overall TKB30 of 98.11% reflects a healthy lending portfolio. Only 1.89% of outstanding balance (\$53M) is classified as DPD>30, well within acceptable industry thresholds.

C2

2012 Cohort is a Structural Anomaly

The 2012 vintage recorded the lowest TKB30 (96.25%), with a 2.62 pp gap vs. the 2019 cohort. This anomaly persists despite better borrower demographics, indicating a systemic pricing issue.

C3

Demographics Do NOT Drive the Anomaly

2012 borrowers had more mortgage ownership (+3.64 pp), longer employment (+0.74 yrs), and higher income (+\$2,000) than other cohorts — ruling out demographic explanations.

C4

Pricing & Loan Mix Are the Root Cause

The 2012 cohort averaged 18.36% interest rates vs. 12.93% for other cohorts (+5.43 pp) and had a 64.29% debt_consolidation share (+9.14 pp) — both well-established delinquency risk factors.

STRATEGIC RECOMMENDATIONS

R1

Implement Rate Cap Policy for High-Risk Segments

Enforce maximum interest rate ceilings for debt consolidation loans. Rates above 16% significantly increase default probability — flag these for enhanced underwriting review.

R2

Strengthen Debt Consolidation Loan Screening

Introduce stricter DTI and income-to-obligation ratio requirements for debt_consolidation applicants. Consider limiting this segment's share in new originations to ≤55%.

R3

Develop Risk-Adjusted Vintage Monitoring

Track TKB30 by issuance year monthly. Establish early-warning thresholds at cohort TKB30 < 97.5% to trigger proactive collection and restructuring initiatives.

R4

Leverage Strong Demographic Profiles

Mortgage holders and long-tenured employees represent lower-risk segments. Create preferential pricing and higher credit limits for these customer profiles to grow quality assets.

R5

Scale Successful 2018–2019 Underwriting Criteria

The 2018–2019 cohorts achieved TKB30 > 98.6% with rates below 12.5%. Codify and operationalize these underwriting standards as the new baseline across all originations.

Thank You

RevoFin Loan Portfolio Analysis — April 2026
